Evaluating Generative Models for Cloud Removal in Satellite Imagery

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Why This Problem Matters

Motivating application: Yunnan and other cloud-prone southern regions

>80%

annual cloud frequency in southern China

Persistent cloud cover renders optical satellites nearly useless

Research question：

"At what cloud coverage threshold do generative models become unreliable?"

Experimental Logic

A controlled benchmark lets us isolate the effect of cloud coverage.

We compare all methods on the **same scenes**, with the **same masks**, and at the **same five occlusion levels**.

01 Ground truth

Cloud-free Sentinel-2 tiles

Used as reference images

02 Synthetic clouds

Masks at 5 levels:

5, 10, 30, 50, 70%

Controlled cloudy inputs

03 Run methods

Temporal

GAN

Diffusion

04 Evaluate

PSNR / SSIM

L1 / cloud-region L1

NDVI MAE

05 Infer boundary

Plot degradation curves

Identify reliability threshold

Performance changes can be attributed directly to increasing cloud coverage rather than to arbitrary scene differences.

Data, Methods, and a Key Limitation

Important context for interpreting the benchmark.

Data

- Sentinel-2, 10 m resolution
- 2,000 cloud-free image tiles
- 70% train / 30% test split
- Input: RGB + NIR
- Output: RGB

Methods

- Temporal compositing baseline
- GAN-based cloud removal
- 70% train / 30% test split
- Diffusion-based inpainting

Training setup: Batch size = 8, Epochs = 5, Learning rate = 0.0002

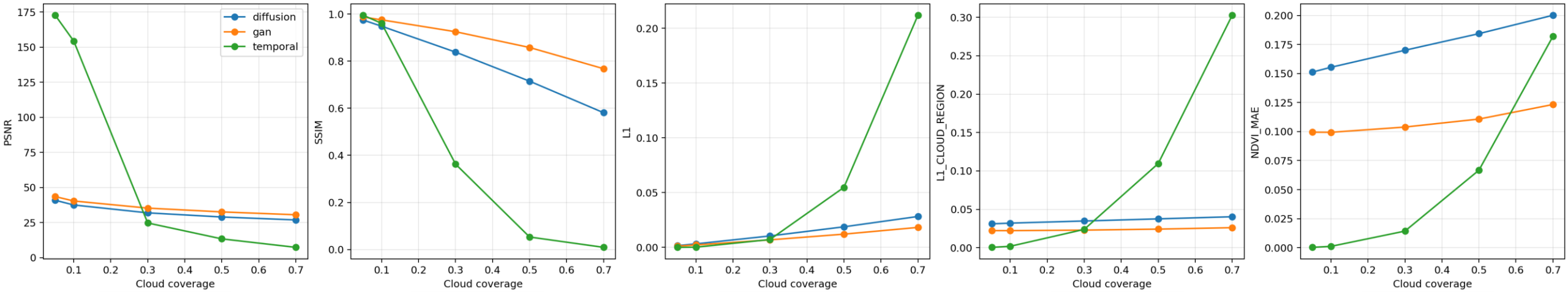
Metrics

- **PSNR**: pixel fidelity (higher is better)
- **SSIM**: structural similarity (higher is better)
- **L1 / cloud-region L1**: reconstruction error (lower is better)
- **NDVI MAE**: vegetation-index error (lower is better)

Important limitation: the data are selected in Beijing, but the motivating application is Yunnan.

This benchmark is stronger for studying occlusion sensitivity than for claiming direct generalization to Yunnan.

Quantitative Comparison Across Cloud Coverage Levels



PSNR **SSIM** **L1** **L1 in cloud region** **NDVI MAE**

Readout 1

GAN is best overall

Best balance of fidelity, structure, and vegetation accuracy as coverage rises.

Readout 2

Temporal is strong only at low coverage

Excellent at 5%–10%, but collapses after ~30% when too much information is missing.

Readout 3

Diffusion underperforms here

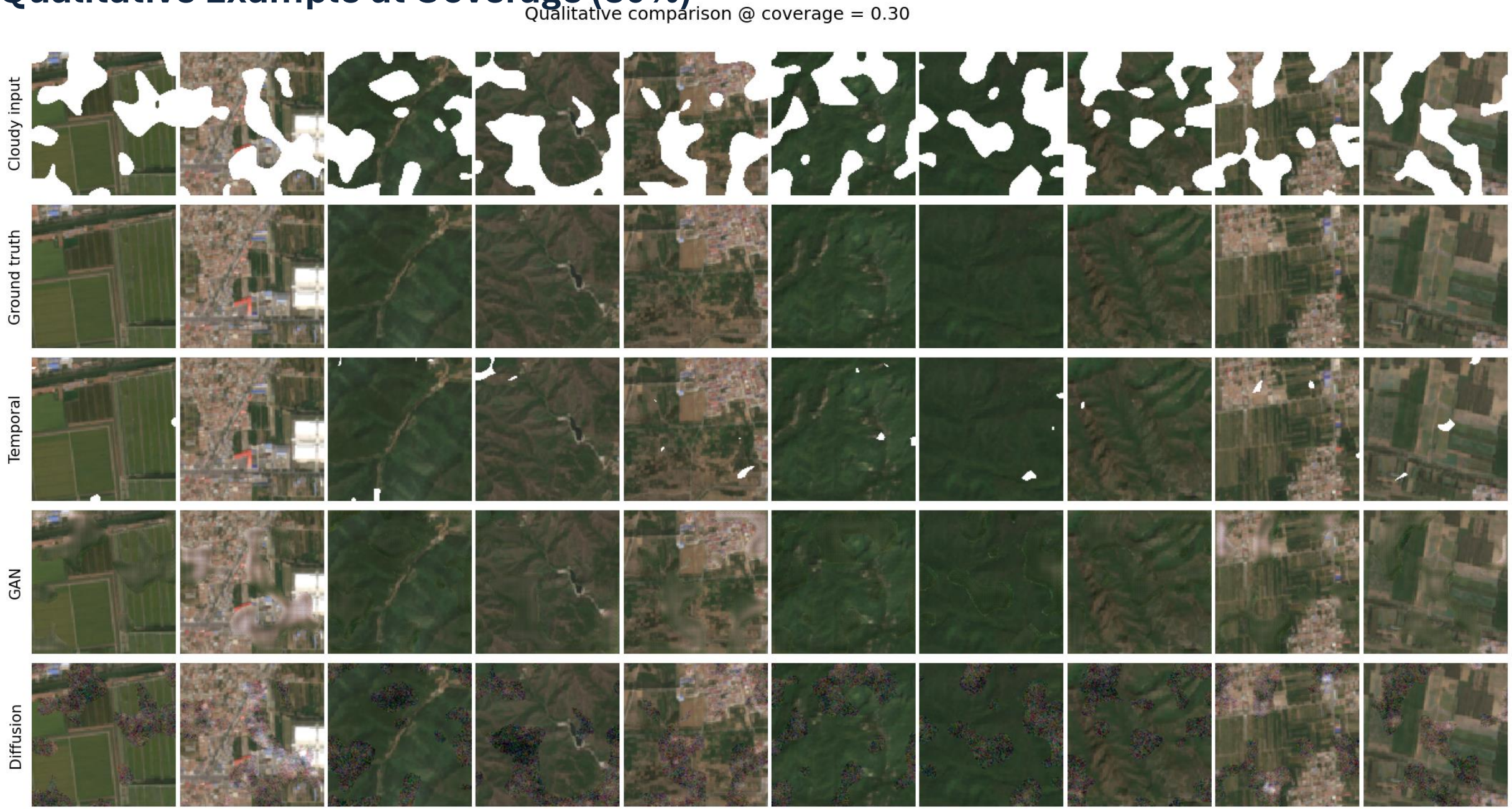
In this setup, it degrades steadily and is not competitive with GAN.

PSNR and SSIM should go up; L1, cloud-region L1, and NDVI MAE should go down.

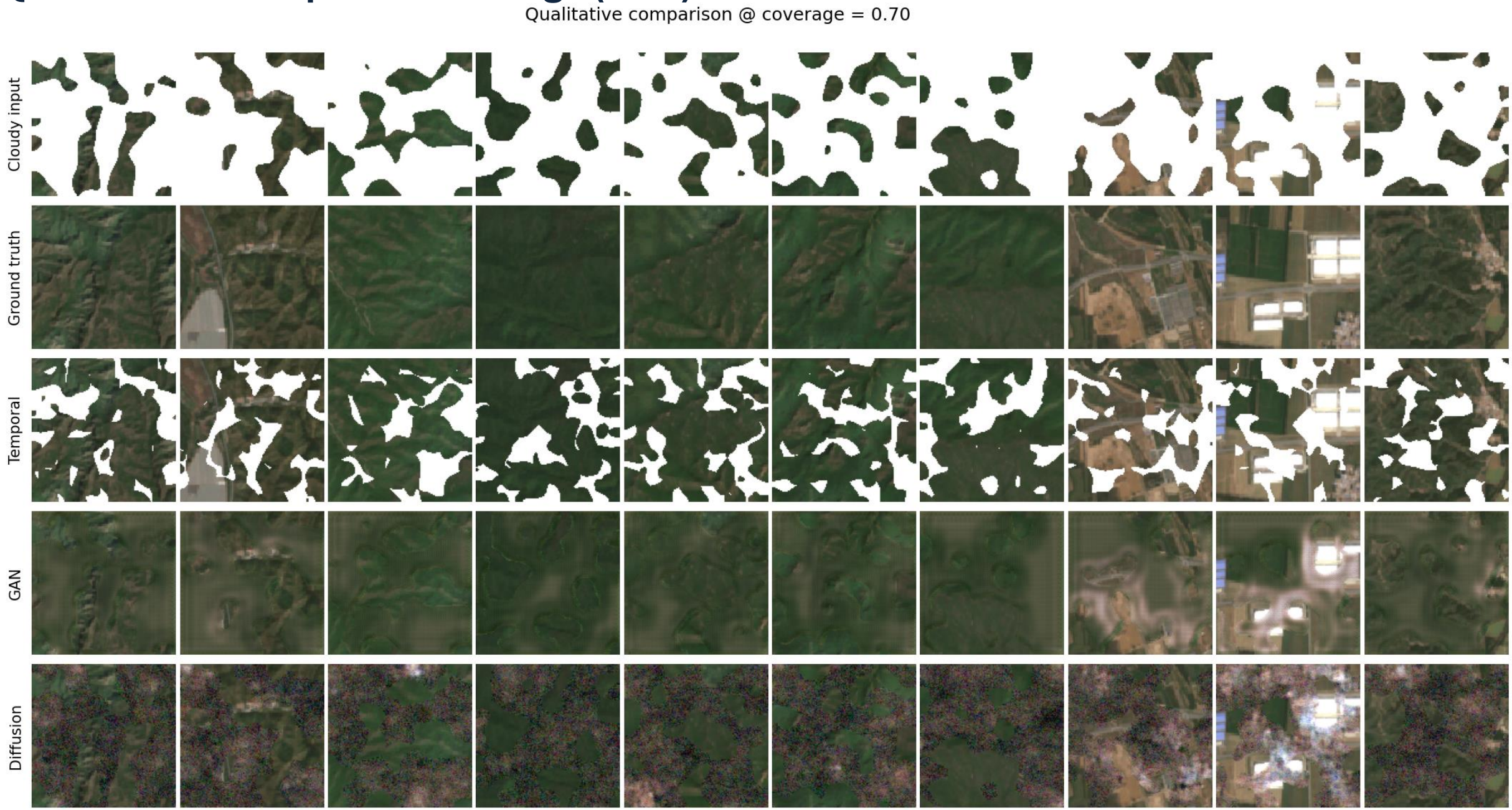
Qualitative Example at Low Coverage (10%)

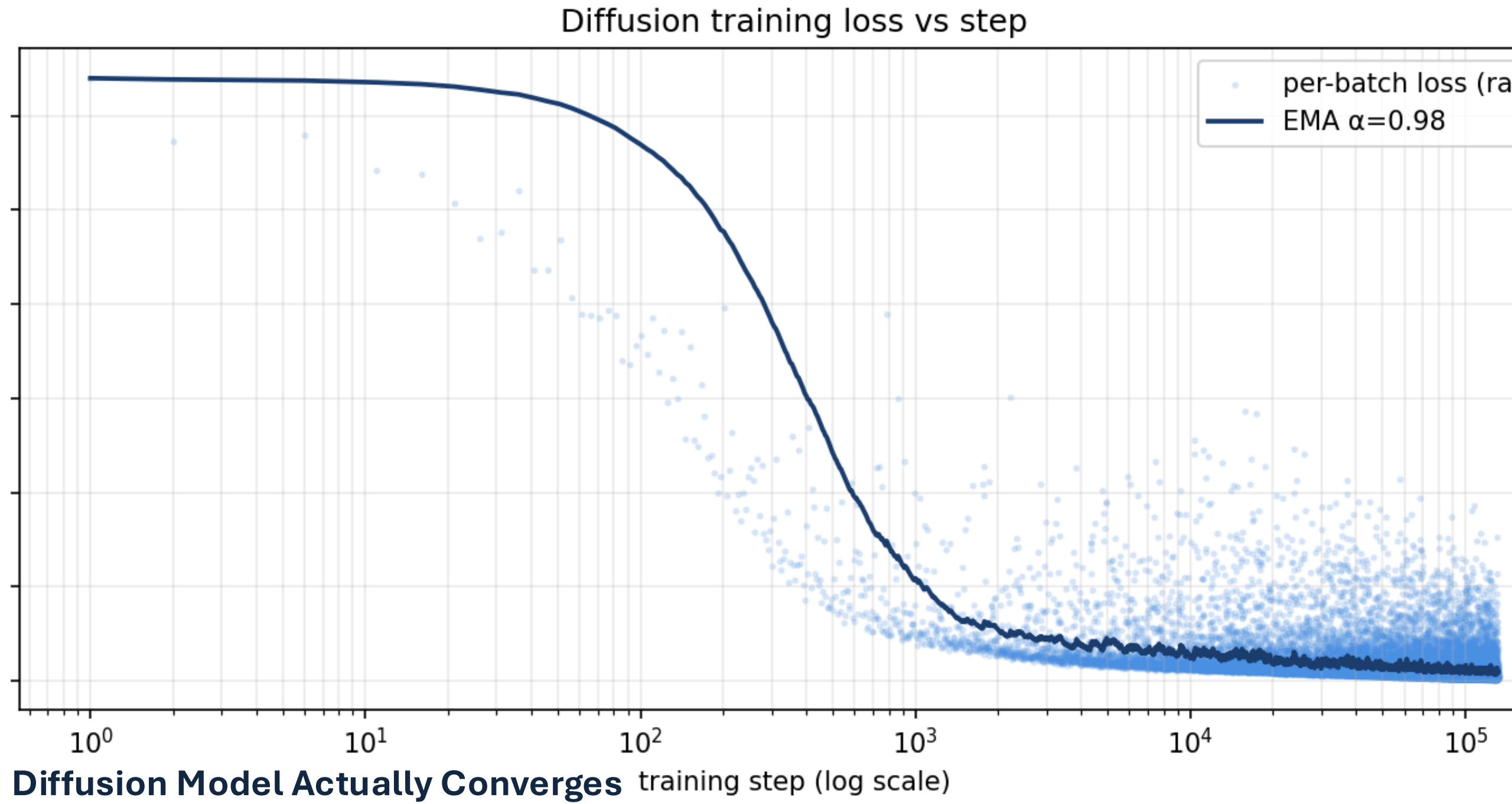
The task is solvable in easier cases

Qualitative Example at Coverage (30%)



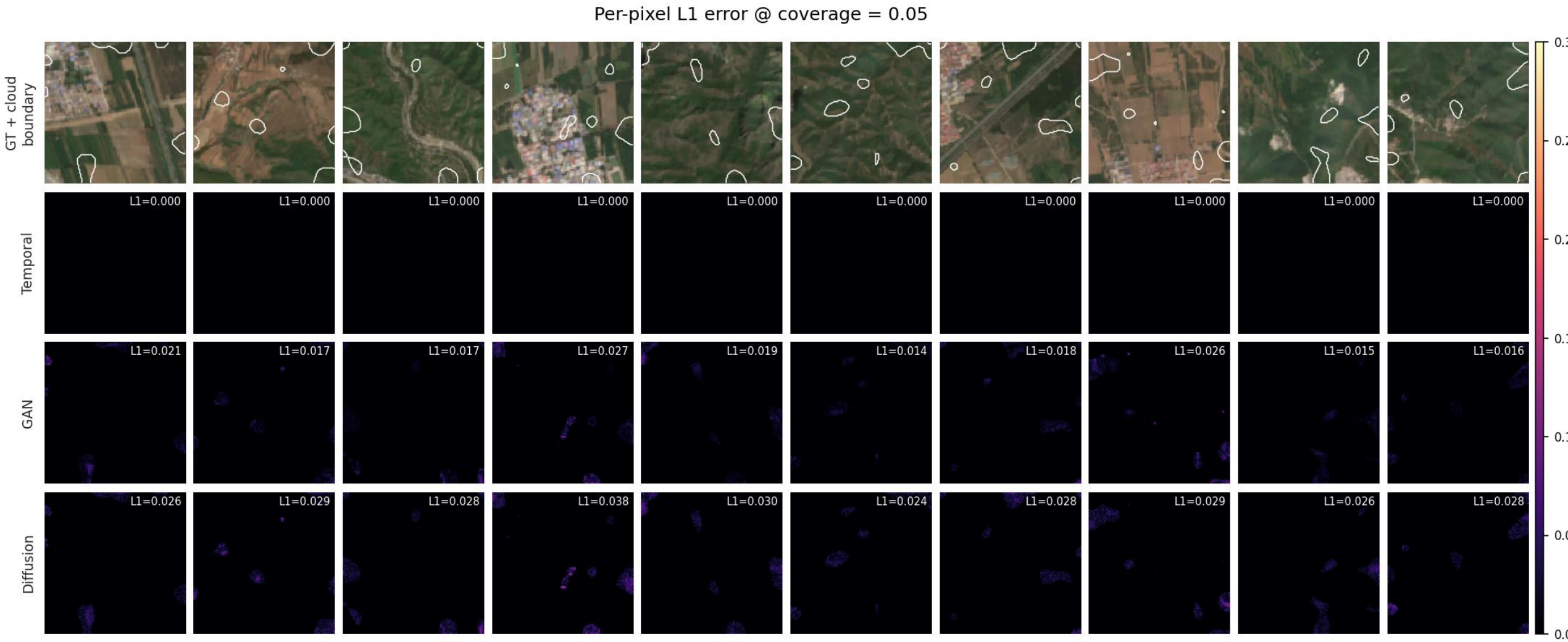
Qualitative Example at Coverage (70%)





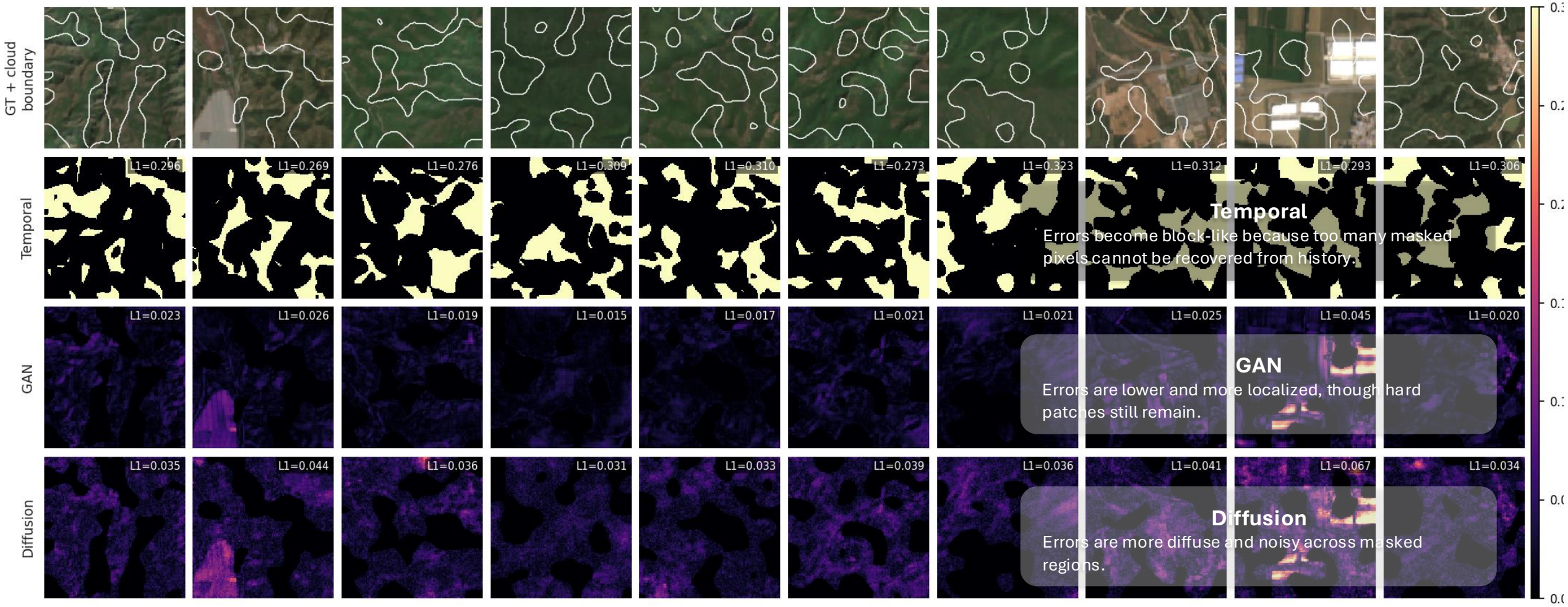
Diffusion Model Actually Converges

Per-pixel L1 Error Heatmap at Coverage (5%)



Per-pixel L1 Error Heatmap at High Coverage (70%)

Hard case: spatial failure modes



The high errors of some patches are particularly concentrated in: White or highly reflective building areas Urban/industrial plots Areas with drastic texture changes. This indicates that under high occlusion, the model is more prone to making incorrect corrections in these complex areas.

Takeaways

Main findings

- Reliability depends strongly on cloud coverage.
- Temporal compositing is excellent at low coverage, but breaks down as missing area grows.
- GAN is the strongest overall method in this benchmark.

Limitations and next steps

- The benchmark uses Beijing tiles, while the motivating application is Yunnan / southern China.
- So the results are more convincing about occlusion sensitivity than about cross-region transfer.
- Next step: test on Yunnan-like or multi-region data to validate whether the same threshold still holds.

Why Did Diffusion Underperform?

1. Limited data and compute: Diffusion models are more data- and compute-intensive than GAN.
2. Simple backbone design: The current backbone (UNet) is too simple for complex remote-sensing textures.
3. Non-targeted optimization: The loss does not emphasize cloud-region recovery enough.
4. Weak mask conditioning: The model does not focus strongly enough on the masked cloud regions.